

SAN MARCOS, TX, USA

WMO: 722539

Lat: 29.891N

Lon: 97.864W

Elev: 597

StdP: 14.38

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 12979

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	27.5	30.2	9.7	9.2	39.3	13.9	11.4	40.0	28.3	49.5	26.2	53.6	8.3	0	0.467

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	22.4	100.0	74.3	98.9	74.4	97.0	74.5	78.5	90.5	78.0	90.0	77.3	89.3	10.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.5	136.5	82.3	75.0	134.2	82.1	73.9	129.0	81.9	42.6	91.0	41.7	90.2	41.0	89.5	83.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
24.6	21.3	19.2	DB	22.3	103.0	3.0	3.1	20.1	105.3	18.4	107.1	16.7	108.8	14.5	111.1
			WB	19.5	80.1	3.1	1.6	17.3	81.3	15.5	82.2	13.8	83.1	11.6	84.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	69.1	50.8	55.3	61.5	69.2	77.1	82.7	84.2	85.1	79.4	70.4	60.6	52.4
	DBStd	14.22	9.12	10.23	9.17	7.43	5.73	3.86	3.33	3.14	5.42	7.91	9.02	9.05
	HDD50	269	101	56	17	1	0	0	0	0	0	1	16	77
	HDD65	1600	446	298	176	42	3	0	0	0	2	40	189	404
	CDD50	7252	126	203	375	577	841	981	1060	1088	883	632	334	152
	CDD65	3107	6	25	68	168	379	531	595	623	435	206	57	14
	CDH74	33821	43	176	478	1421	3652	6141	7185	7844	4549	1873	388	71
	CDH80	16617	3	36	103	453	1576	3150	3907	4411	2238	671	65	4
Wind	WSAvg	9.0	9.5	10.0	10.6	10.7	9.9	9.0	8.0	7.6	7.1	8.0	8.6	8.9
Precipitation	PrecAvg	35.7	2.2	2.3	1.7	2.8	5.0	4.4	1.9	2.4	3.6	3.3	3.2	2.1
	PrecMax	49.6	8.2	7.3	5.2	8.1	13.1	23.0	9.1	7.3	7.4	12.4	13.0	7.6
	PrecMin	19.9	0.1	0.3	0.0	0.2	0.3	0.0	0.0	0.3	0.4	0.0	0.2	0.2
	PrecStd	8.6	1.8	1.6	1.2	2.1	3.5	4.3	2.2	2.1	2.1	2.8	2.9	1.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	79.4	83.9	87.5	92.9	97.2	100.3	101.5	102.3	101.8	92.9	85.0	79.9
		MCWB	62.1	63.8	66.3	69.2	74.9	74.5	74.8	74.4	72.0	73.2	70.6	65.3
	2%	DB	74.3	79.3	82.3	88.4	93.4	98.7	99.5	100.1	97.2	90.1	81.6	75.3
		MCWB	60.9	62.9	66.1	69.4	74.0	74.5	74.4	74.8	73.4	72.9	67.8	63.0
	5%	DB	71.7	74.9	79.4	84.4	91.0	95.8	97.8	99.0	93.5	87.7	79.0	72.5
		MCWB	60.4	61.9	65.2	68.4	73.7	75.4	74.3	74.7	73.4	71.4	65.8	62.2
	10%	DB	67.5	71.8	75.3	81.8	89.2	93.0	95.4	97.0	91.0	83.7	75.0	69.6
		MCWB	58.9	61.8	63.6	67.2	73.7	74.8	74.7	75.0	73.2	70.3	65.0	61.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.7	69.3	71.7	75.1	78.9	80.3	79.1	79.1	78.5	77.8	73.2	69.4
		MCDB	74.2	75.5	80.7	85.6	91.1	90.8	92.3	93.7	89.5	86.3	80.7	75.6
	2%	WB	65.7	67.7	69.7	73.2	77.4	78.6	78.1	78.1	77.6	76.2	71.2	67.5
		MCDB	71.4	74.2	77.9	83.0	89.0	89.5	90.9	91.8	88.6	84.4	78.1	72.8
	5%	WB	63.3	66.1	68.1	71.6	76.2	77.8	77.4	77.4	76.8	74.8	69.3	65.7
		MCDB	68.0	71.6	75.5	80.4	86.6	89.1	89.7	90.5	87.4	82.6	76.0	71.0
	10%	WB	60.1	63.9	66.5	70.2	74.9	76.9	76.7	76.8	75.8	73.2	67.4	62.4
		MCDB	66.1	69.1	73.5	78.7	84.6	88.0	88.7	89.8	85.6	80.4	73.9	67.2
Mean Daily Temperature Range	5% DB	MDBR	22.4	22.0	22.0	22.6	20.3	20.4	21.1	22.4	21.3	22.1	22.0	22.4
		MCDBR	27.7	26.5	24.1	24.6	22.7	23.0	24.3	25.0	24.4	23.7	23.6	26.7
		MCWBR	18.5	16.1	13.5	11.9	8.6	6.8	6.3	6.8	8.7	10.9	13.8	17.9
	5% WB	MCDBR	21.5	19.7	20.0	17.8	19.3	19.1	20.4	21.8	19.6	18.9	19.0	21.0
		MCWBR	16.7	13.7	12.5	9.9	8.5	7.1	6.8	6.7	7.9	10.2	12.1	16.5
Clear-Sky Solar Irradiance	taub	0.337	0.337	0.363	0.402	0.449	0.430	0.443	0.429	0.415	0.375	0.362	0.320	
	taud	2.466	2.480	2.418	2.325	2.240	2.344	2.340	2.363	2.358	2.451	2.425	2.532	
	Ebn at Noon	284	295	293	283	269	272	268	272	272	278	272	285	
	Edh at Noon	30	32	36	41	45	40	40	39	38	32	30	26	
All-Sky Solar Radiation	RadAvg	953	1106	1421	1701	1833	2087	2101	1993	1643	1364	1024	844	
	RadStd	104	198	133	156	163	134	175	108	156	182	116	108	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (39 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+148

Nomenclature: See separate page